

1994 HKCEE MATHS Paper II

1 If $f(x) = x^2 + 2x$, then $f(x-1) =$

A. x^2

B. $x^2 - 1$

D. $x^2 + 2x - 3$

C. $x^2 + 2x - 1$

E. $x^2 + 4x - 1$

2 If $y = \frac{2x-1}{x+2}$, then $x =$

A. $\frac{1+3y}{2}$

B. $\frac{1+2y}{2+y}$

D. $\frac{1-2y}{2+y}$

C. $\frac{1+2y}{2-y}$

E. $\frac{1-2y}{2-y}$

3 The L.C.M. of $(x-1)^2$, $x^2 - 1$ and $x^3 - 1$ is

A. $x-1$

B. $(x-1)^4(x+1)(x^2+x+1)$

C. $(x-1)^2(x+1)(x^2+x+1)$

D. $(x-1)^2(x+1)(x^2-x+1)$

E. $(x-1)(x+1)(x^2+x+1)$

4 If $a = \sqrt{3} + \sqrt{2}$, then $a - \frac{1}{a} =$

A. 0

B. $2\sqrt{2}$

D. $\sqrt{3} - \sqrt{2}$

C. $2\sqrt{3}$

E. $\frac{2\sqrt{3}}{3} + \frac{\sqrt{2}}{2}$

5 In the figure, (x, y) is a point in the shaded region (including the boundary) and x, y are integers.Find the greatest value of $3x + y$.

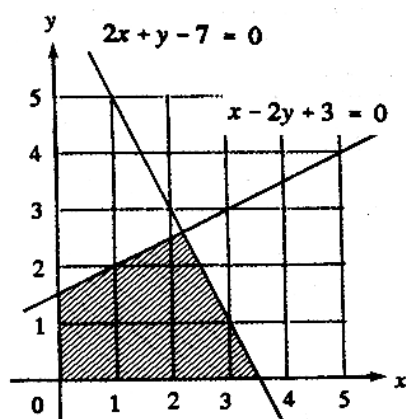
A. 7

B. 8

C. 9.2

D. 10

E. 10.3



6 If $x(x+1) < 5(x+1)$, then

A. $x \leq 5$

B. $x < -5$ or $x > 1$

C. $x \leq -1$ or $x > 3$

D. $-5 < x < 1$

E. $-1 < x < 3$

7 Which of the following is/are an identity/identities?

Label 1 $(x+2)(x-2) = x^2 - 4$

Label 1 $(x+2)(x-2) = 0$

Label 1 $(x+2)^3 = x^3 + 8$

A. I only

B. II only

D. I and III only

C. III only

E. II and III only

8 If $\alpha \neq \beta$ and $\begin{cases} 3\alpha^2 - h\alpha - b = 0 \\ 3\beta^2 - h\beta - b = 0 \end{cases}$ Figure 1, then $\alpha + \beta =$

A. $-\frac{b}{3}$

B. $\frac{b}{3}$

D. $-\frac{h}{3}$

C. $\frac{h}{3}$

E. $\frac{h}{3}$

9 Mr. Chan bought a car for \$143 900. If the value of the car goes down by 10% each year, find its value at the end of the third year. (Give your answer correct to the nearest hundred dollars.)

A. \$94 400

B. \$100 700

C. \$104 900

D. \$115 100

9. \$116 600
10. A wholesaler sells an article to a retailer at a profit of 20%. The retailer sells it to a customer for \$3 600 at a profit of \$720. Find the original cost of the article to the wholesaler.

- A. \$2 304
B. \$2 400
C. \$2 880
D. \$3 000
E. \$3 456

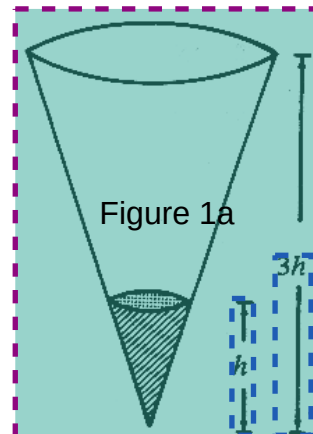
11. The bearing of A from B is 075° . What is the bearing of B from A ?

- A. 015°
B. 075°
C. 105°
D. 195°
E. 255°

12. If the sum to infinity of a G.S. is $\frac{81}{4}$ and its second term is -9 , the common ratio is

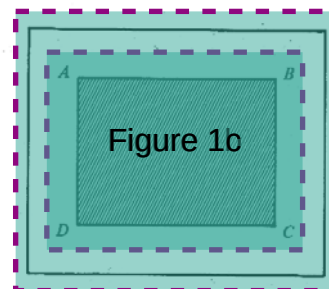
- A. $-\frac{1}{3}$
B. $\frac{1}{3}$
C. $-\frac{4}{3}$
D. $\frac{4}{3}$
E. $-\frac{4}{9}$

13. In the figure, the paper cup in the form of a circular cone contains 10ml of water. How many ml of water must be added to fill up the paper cup?



- A. 20
B. 80
C. 90
D. 260
E. 270

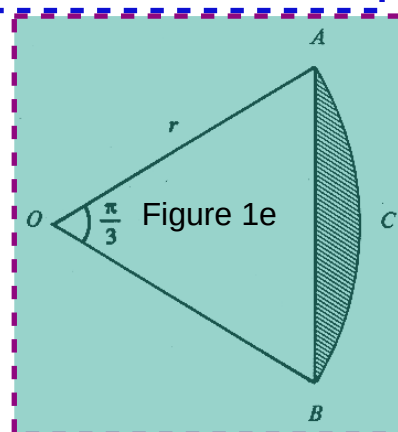
14. In the figure, $ABCD$ is a rectangular field of length p metres and width q metres. The path around the field is of width 2 metres. Find the area of the path.



- A. $(4p + 4q)m^2$
B. $(2p + 2q + 4)m^2$
C. $(2p + 2q + 16)m^2$
D. $(4p + 4q + 16)m^2$
E. $(pq + 4p + 4q + 16)m^2$

15. In the figure, $OACB$ is a sector of radius r . If $\angle AOB = \frac{\pi}{3}$, find the area of the shaded part.

- A. $\left(\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right)r^2$
B. $\left(\frac{\pi}{6} - \frac{1}{4}\right)r^2$
C. $\left(\frac{\pi}{3} - \frac{\sqrt{3}}{2}\right)r^2$



D. $\left(\frac{\pi}{3} - \frac{1}{2}\right)r^2$

E. $\frac{\pi}{3}r - \frac{\sqrt{3}}{4}r^2$

16. $\frac{\cos \theta}{\sin \theta + 1} - \frac{\cos \theta}{\sin \theta - 1} =$

A. $\frac{2}{\cos \theta}$

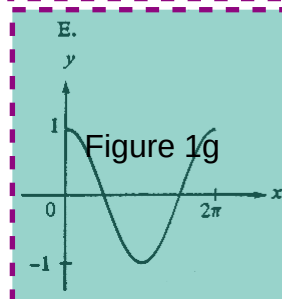
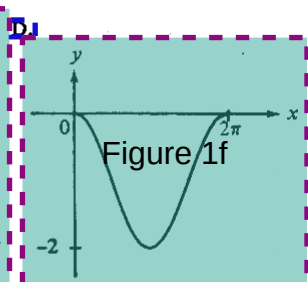
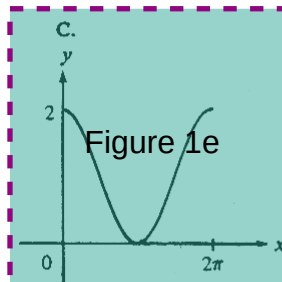
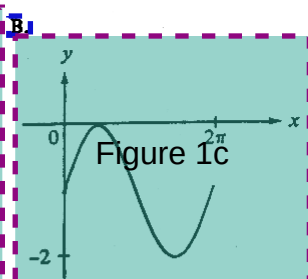
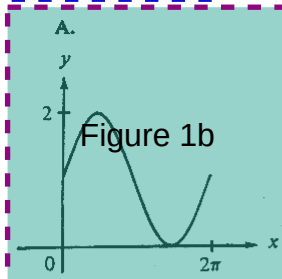
B. $\frac{2}{\cos \theta}$

C. 0

D. $2 \tan \theta$

E. $-2 \tan \theta$

17. Which of the following figures shows the graph of $y = 1 + \sin x$?



18. $\frac{\sin(180^\circ + \theta)}{\cos(90^\circ - \theta)} =$

A. $\tan \theta$

19. In the figure, $ABCD$ is a cyclic quadrilateral with $AB=5$, $BC=2$ and $\angle ADC=120^\circ$. Find AC .

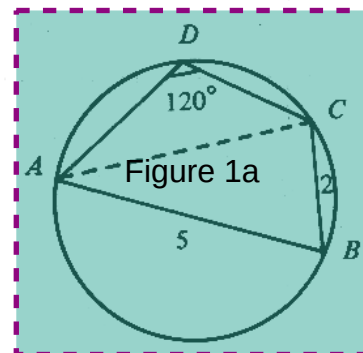
A. $\sqrt{19}$

B. $\sqrt{21}$

C. $2\sqrt{6}$

D. $\sqrt{34}$

E. $\sqrt{39}$



20. In the figure, PC is a vertical pole standing on the horizontal plane ABC . If $\angle ABC=90^\circ$, $\angle BAC=30^\circ$, $AC=6$ and $PC=5$, find $\tan \theta$.

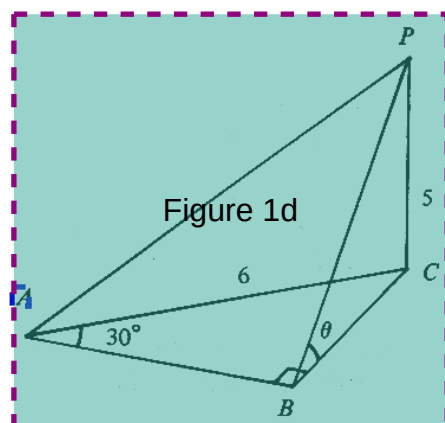
A. $\frac{3}{5}$

B. $\frac{5}{6}$

C. $\frac{5}{3}$

D. $\frac{3\sqrt{3}}{5}$

E. $\frac{5\sqrt{3}}{9}$

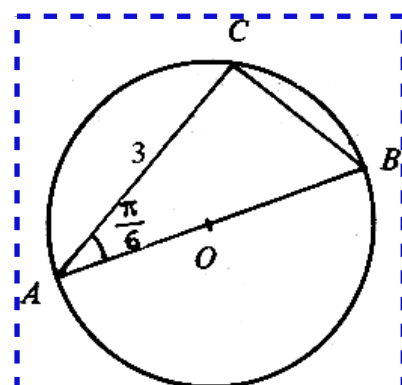


21. In the figure, O is the center of the circle. If $AC=3$ and $\angle BAC=\frac{\pi}{6}$, find the diameter AB .

A. $\frac{3}{2}$

B. 6

C. $\frac{3\sqrt{3}}{2}$



D. $2\sqrt{3}$

E. $3\sqrt{3}$

- 22 In the figure, PA is tangent to the circle at A . $\angle CAP = 28^\circ$ and $BA = BC$. Find x .

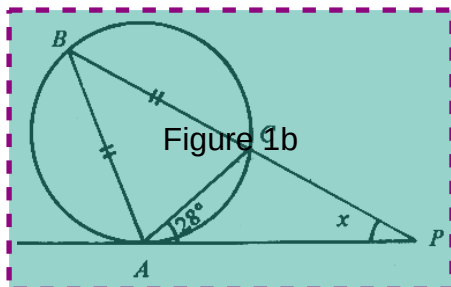
A. 28°

B. 48°

C. 56°

D. 62°

E. 76°



- 23 In the figure, O is the center of the inscribed circle of $\triangle ABC$. If $\angle OAC = 30^\circ$ and $\angle OCA = 25^\circ$, find $\angle ABC$.

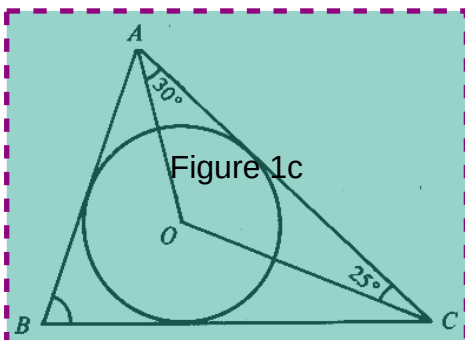
A. 50°

B. 55°

C. 60°

D. 62.5°

E. 70°



- 24 In the figure, $AB = AD$ and $BC = CD$. If $\angle BAD = 80^\circ$ and $\angle ADC = 65^\circ$, then $\angle BCD =$

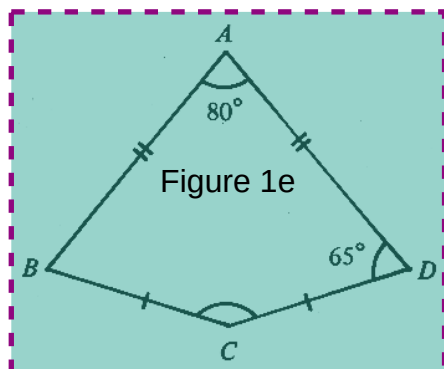
A. 100°

B. 130°

C. 145°

D. 150°

E. 160°



- 25 In the figure, x , y and z are the exterior angles of $\triangle ABC$. If $x : y : z = 4 : 5 : 6$, then $\angle BAC =$

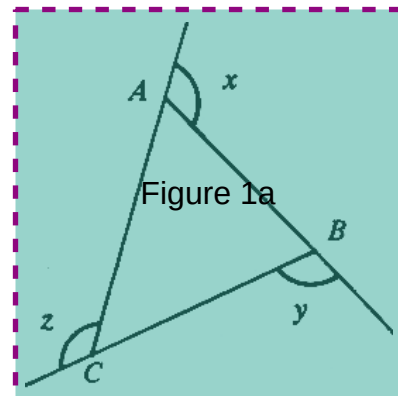
A. 48°

B. 84°

C. 96°

D. 120°

E. 132°



- 26 The points $A(4, -1)$, $B(-2, 3)$ and $C(x, 5)$ lie on a straight line. Find x .

A. -5

B. -4

C. 0

D. 2

E. 3

- 27 In the figure, the shaded part is bounded by the axes, the lines $x = 3$ and $x + y = 5$. Find its area.

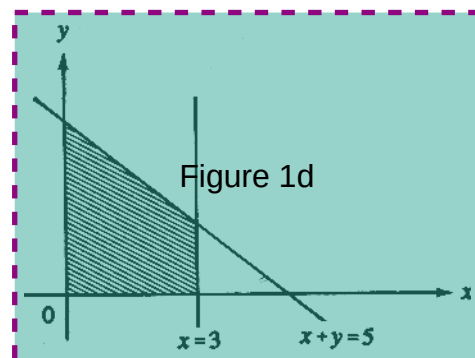
A. 10.5

B. 12

C. 13

D. 19.5

E. 21



- 28 AB is a diameter of the circle $x^2 + y^2 - 2x - 2y - 18 = 0$. If A is $(3, 5)$, then B is

A. $(2, 3)$

- B. $(1, -1)$ D. $(-5, -7)$
 C. $(-1, -3)$ E. $(-7, -9)$

29 The equations of two circles

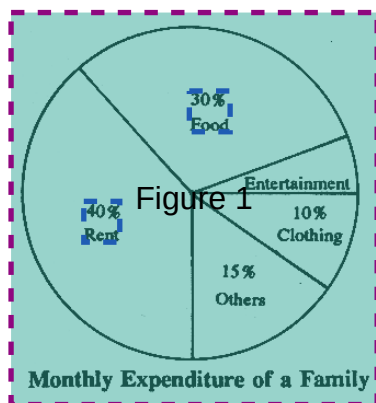
$$\text{are } \begin{cases} x^2 + y^2 - 4x - 6y = 0 \\ x^2 + y^2 + 4x + 6y = 0 \end{cases}$$

Which of the following is/are true?

- I. The two circles have the same center.
 II. The two circles have equal radii.
 III. The two circles pass through the origin.

- A. I only
 B. II only D. I and III only
 C. III only E. II and III only
- 30 In the figure, the pie chart shows the monthly expenditure of a family. If the family spends \$4800 monthly on rent, what is the monthly expenditure on entertainment?

- A. \$240
 B. \$600
 C. \$720
 D. \$1800
 E. \$12 000



31 A box contains 5 eggs, 2 of which are rotten. If 2 eggs are chosen at random, find the probability that exactly one of them is rotten.

- A. $\frac{2}{5}$
 B. $\frac{3}{5}$ D. $\frac{6}{25}$

32 The mean, standard deviation and interquartile range of n numbers are m , s and q respectively. If 3 is added to each of the n numbers, what will be their new mean, standard deviation and interquartile range?

	Mean	Standard Deviation	Interquartile Range
A.	m	s	q
B.	m	$s + 3$	$q + 3$
C.	$m + 3$	s	q
D.	$m + 3$	s	$q + 3$
E.	$m + 3$	$s + 3$	$q + 3$

33 $(3^x)^2 =$

- A. $3^{(x^2)}$
 B. 3^{x+2} D. 6^x
 C. 3^{2x} E. 9^{2x}

34 If $\log 2 = a$ and $\log 9 = b$, then $\log 12 =$

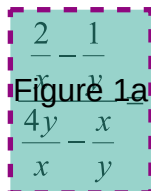
- A. $2a + \frac{b}{3}$
 B. $2a + \frac{b}{2}$ D. $a^2 + b^2$
 C. $\frac{2}{3}a + \frac{2}{3}b$ E. $a^2 b^2$

35 Factorize $a^2 - 2ab + b^2 - a + b$.

- A. $(a-b)(a-b-1)$
 B. $(a-b)(a-b+1)$

- C. $(a-b)(a+b-1)$
 D. $(a+b)(a-b+1)$
 E. $(a-b-1)^2$

36



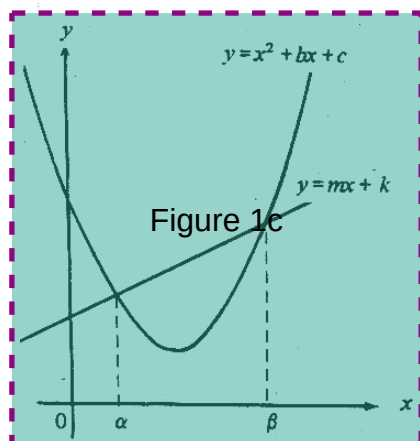
- A. $2y-x$
 B. $2y+x$
 C. $\frac{1}{2y-x}$
 D. $\frac{1}{2y+x}$
 E. $\frac{1}{4y-x}$

37. $P(x)$ is a polynomial. When $P(x)$ is divided by $(5x-2)$, the remainder is R . If $P(x)$ is divided by $(2-5x)$, then the remainder is

- A. R
 B. $-R$
 C. $\frac{2}{5}R$
 D. $\frac{2}{5}$
 E. $-\frac{2}{5}$

38. In the figure, the line $y = mx + k$ cuts the curve $y = x^2 + bx + c$ at $x = \alpha$ and $x = \beta$. Find the value of $\alpha\beta$.

- A. $c-b$
 B. α
 C. $m-b$
 D. $k-c$
 E. $c-k$



39. If $x = 3$, $y = 2$ satisfy the simultaneous equations $\begin{cases} ax + by = 2 \\ bx - ay = 3 \end{cases}$, find the values of a and b .

- A. $a = 0$, $b = 1$
 B. $a = 0$, $b = -1$
 C. $a = \frac{5}{6}$, $b = -\frac{1}{4}$
 D. $a = -\frac{1}{13}$, $b = \frac{37}{39}$
 E. $a = -\frac{12}{13}$, $b = \frac{5}{13}$

40. From the table, which of the following intervals must contain a root of $f(x) - x = 0$?

x	$f(x)$
-2	1.2
-1	0.8
0	0.7
1	0.2
2	-0.1
3	0.8

- A. $-2 < x < -1$
 B. $-1 < x < 0$
 C. $0 < x < 1$
 D. $1 < x < 2$
 E. $2 < x < 3$
41. If the product of the first n terms of the sequence $10, 10^2, 10^3, \dots, 10^n$ exceeds 10^{55} , find the minimum value of n .

- A. 9
 B. 10
 C. 11
 D. 12
 E. 36

- 42 If $a:b=2:3$, $a:c=3:4$ and $a:d=4:5$, then $b:c:d=$

A. $2:3:4$

B. $3:4:5$

D. $18:16:15$

C. $3:6:10$

E. $40:45:48$

- 43 Let x vary inversely as $\sqrt{y}$. If y is increased by 69%, then x will be

A. increased by 23.1%(3 sig. fig.)

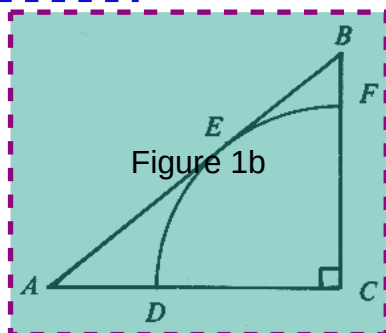
B. increased by 30%

C. decreased by 23.1%(3 sig. fig.)

D. decreased by 30%

E. decreased by 76.9%(3 sig. fig.)

- 44 In the figure, $CDEF$ is a sector of a circle which touches AB at E . If $AB=25$ and $BC=15$, find the radius of the sector.



A. 9

B. 10

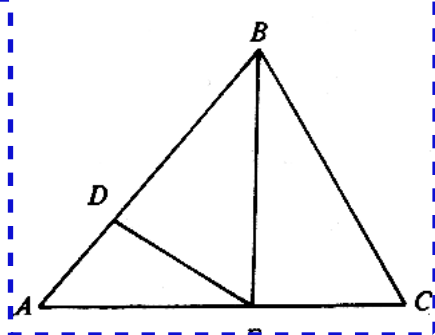
D. 12

C. 12.5

E. 12.5

- 45 In the figure, $AD:DB=1:2$, $AE:EC=3:2$.

Area of $\triangle BDE$: Area of $\triangle ABC =$



A. $1:3$

B. $2:5$

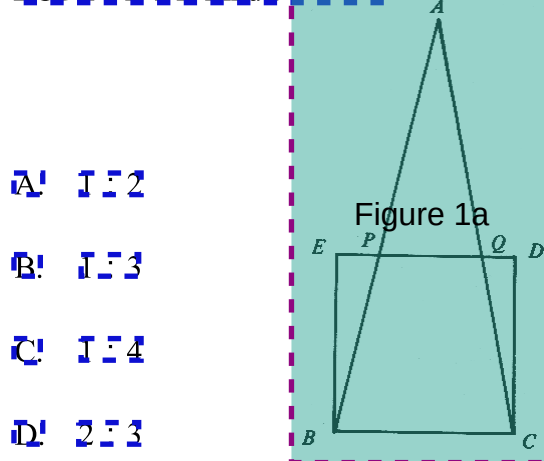
D. $4:25$

C. $3:4$

E. $36:65$

- 46 In the figure, area of $\triangle ABC$: area of square $BCDE = 2:1$. Find $PE:PC$.

A. $1:2$



A. $1:2$

B. $1:3$

C. $1:4$

D. $2:3$

E. $3:4$

- 47 For $0^\circ \leq x \leq 360^\circ$, how many roots does the equation $\sin x(\cos x + 2) = 0$ have?

A. 0

B. 1

D. 3

C. 2

E. 4

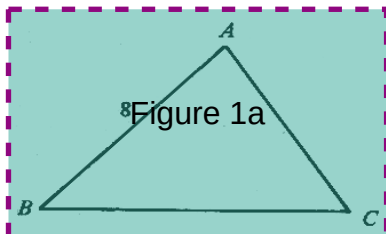
- 48 The largest value of $(3 \cos 2\theta - 1)^2 + 1$ is

A. 2

- B. 5 D. 26
C. 17 E. 50

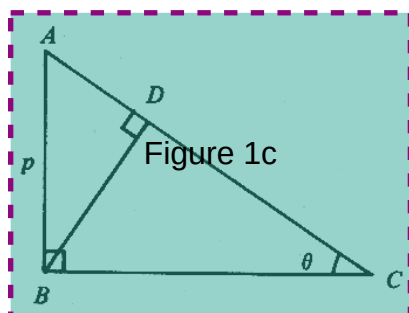
- 49 In the figure, $\sin A : \sin B : \sin C = 4 : 5 : 6$.
If $AB=8$, find AC .

- A. $5\frac{1}{3}$
B. $6\frac{2}{3}$
C. $9\frac{3}{5}$
D. 10
E. 12

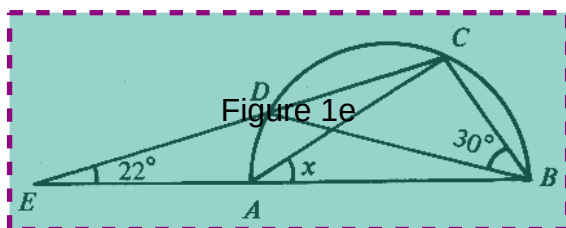


- 50 In the figure, $AB=p$, $\angle ACB=\theta$. Find CD .

- A. $p \sin \theta$
B. $p \cos \theta$
C. $\frac{p \sin \theta}{\cos^2 \theta}$
D. $\frac{p \sin^2 \theta}{\cos \theta}$
E. $\frac{p \cos^2 \theta}{\sin \theta}$

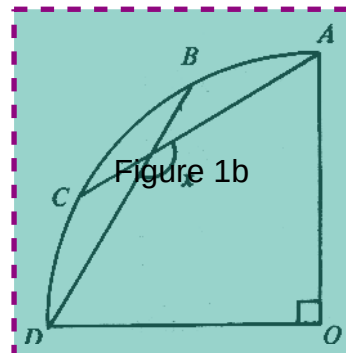


- 51 In the figure, $ABCD$ is a semi-circle, CDE and BAE are straight lines. If $\angle CBD=30^\circ$ and $\angle DEA=22^\circ$, find x .

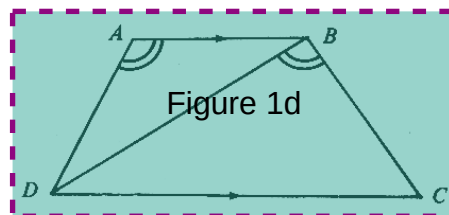


- 52 In the figure, $OABCD$ is a sector of a circle. If $\widehat{AB} = \widehat{BC} = \widehat{CD}$, then $x =$

- A. 105°
B. 120°
C. 135°
D. 144°
E. 150°



- 53 In the figure, $AB \parallel DC$ and $\angle DAB = \angle DBC$.
Which of the following is/are true?



Label I $\frac{AB}{BD} = \frac{BD}{DC}$

Label II $\frac{AB}{BD} = \frac{AD}{BC}$

Label III $\frac{AD}{BD} = \frac{BD}{CD}$

- A. I only
B. II only
C. III only
D. I and II only
E. II and III only